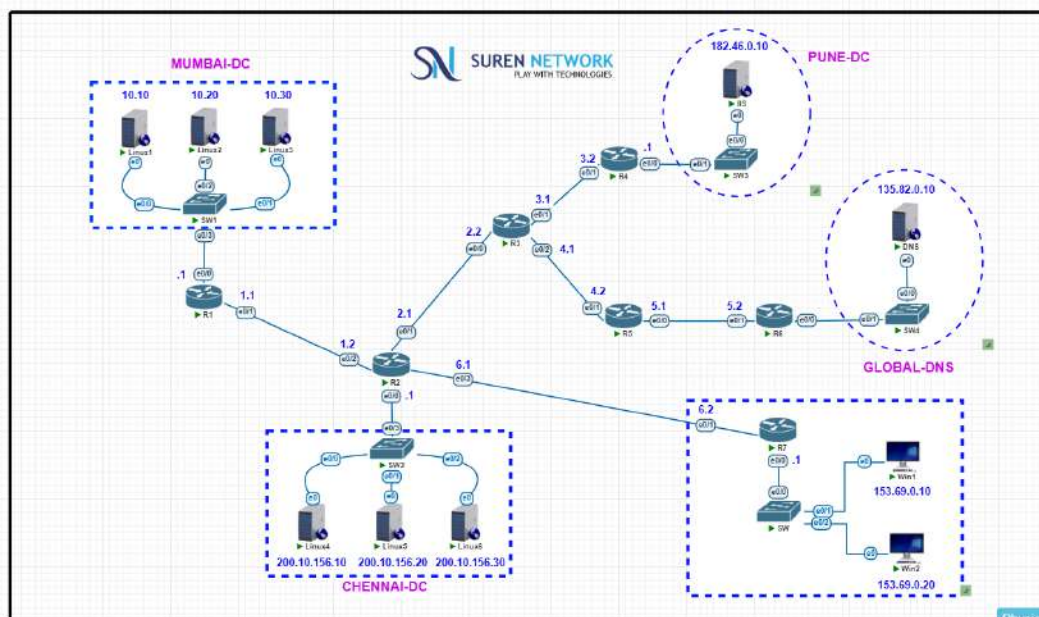




R1

```
R1(config)#
R1(config)#
R1(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 1.0.0.2 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 1.0.0.2
      1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      1.0.0.0/8 is directly connected, Ethernet0/1
L      1.0.0.1/32 is directly connected, Ethernet0/1
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      10.0.0.0/8 is directly connected, Ethernet0/0
L      10.0.0.1/32 is directly connected, Ethernet0/0
R1(config)#
```

R2

```
R2#
R2#
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#show ip route
^
% Invalid input detected at '^' marker.

R2(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       1.0.0.0/8 is directly connected, Ethernet0/2
L       1.0.0.2/32 is directly connected, Ethernet0/2
2.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       2.0.0.0/8 is directly connected, Ethernet0/1
L       2.0.0.1/32 is directly connected, Ethernet0/1
S       3.0.0.0/8 [1/0] via 2.0.0.2
S       4.0.0.0/8 [1/0] via 2.0.0.2
S       5.0.0.0/8 [1/0] via 2.0.0.2
C       6.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       6.0.0.0/8 is directly connected, Ethernet0/3
L       6.0.0.1/32 is directly connected, Ethernet0/3
S       10.0.0.0/8 [1/0] via 1.0.0.1
S       135.82.0.0/16 [1/0] via 2.0.0.2
S       153.69.0.0/16 [1/0] via 6.0.0.2
S       182.46.0.0/16 [1/0] via 2.0.0.2
C       200.10.156.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.10.156.0/24 is directly connected, Ethernet0/0
L       200.10.156.1/32 is directly connected, Ethernet0/0
R2(config)#
R2(config)#
R2(config)#
```

R3

```
R3#
R3#
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR
```

Gateway of last resort is not set

```
S    1.0.0.0/8 [1/0] via 2.0.0.1
C    2.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L    2.0.0.0/8 is directly connected, Ethernet0/0
L    2.0.0.2/32 is directly connected, Ethernet0/0
C    3.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    3.0.0.0/8 is directly connected, Ethernet0/1
L    3.0.0.1/32 is directly connected, Ethernet0/1
C    4.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    4.0.0.0/8 is directly connected, Ethernet0/2
L    4.0.0.1/32 is directly connected, Ethernet0/2
S    5.0.0.0/8 [1/0] via 4.0.0.2
S    6.0.0.0/8 [1/0] via 2.0.0.1
S    10.0.0.0/8 [1/0] via 2.0.0.1
S    135.82.0.0/16 [1/0] via 4.0.0.2
S    153.69.0.0/16 [1/0] via 2.0.0.1
S    200.0.0.0/24 [1/0] via 2.0.0.1
```

```
R3#
R3#
R3#
R3#
R3#
```



QEMU (Linux4)



```
shutting down loopback interface lo ..... [ OK ]
bringing up loopback interface lo ..... [ OK ]
bringing up Ethernet interface eth0 ..... [ OK ]
set up default gateway ..... [ OK ]
#
#
# ping 10.0.0.10
PING 10.0.0.10 (10.0.0.10): 56 data bytes
64 bytes from 10.0.0.10: seq=0 ttl=62 time=8.281 ms
64 bytes from 10.0.0.10: seq=1 ttl=62 time=2.508 ms
64 bytes from 10.0.0.10: seq=2 ttl=62 time=2.670 ms
64 bytes from 10.0.0.10: seq=3 ttl=62 time=2.561 ms
64 bytes from 10.0.0.10: seq=4 ttl=62 time=1.983 ms
64 bytes from 10.0.0.10: seq=5 ttl=62 time=2.014 ms
64 bytes from 10.0.0.10: seq=6 ttl=62 time=2.439 ms
64 bytes from 10.0.0.10: seq=7 ttl=62 time=3.155 ms
64 bytes from 10.0.0.10: seq=8 ttl=62 time=3.993 ms
64 bytes from 10.0.0.10: seq=9 ttl=62 time=3.353 ms
64 bytes from 10.0.0.10: seq=10 ttl=62 time=3.474 ms
64 bytes from 10.0.0.10: seq=11 ttl=62 time=2.776 ms

--- 10.0.0.10 ping statistics ---
12 packets transmitted, 12 packets received, 0% packet loss
round-trip min/avg/max = 1.983/3.267/8.281 ms
#
```

